A Highly Clean Recipe Dataset with Ingredient States Annotation for State Probing Task

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Abstract

Large language models (LLMs) trained on procedural texts rarely encounter explicit descriptions of real-world state changes; in cooking recipes, intermediate ingredient states are almost always omitted, making implicit state tracking particularly difficult. Existing recipe benchmarks focus primarily on surface-level tasks such as step ordering and recipe–image alignment, leaving the deeper challenge of modeling step-wise world-state transitions largely unexplored. We introduce the **NHK Recipe State Probing** benchmark, built on a professionally curated subset of approximately 25,000 Japanese recipes from *Minna no Kyo no Ryori* (NHK Educational). Our core contribution is a fine-grained annotation of 100 recipes with explicit per-step *ingredient state graphs*, capturing how ingredients transform throughout the cooking process. From this annotation we construct three novel probing tasks: (1) **Step Reference Resolution**, which tests whether a model can identify which prior step is referenced within a recipe step; (2) **Ingredient Usage Prediction**, which asks whether a given ingredient remains unused up to a specified step; and (3) **Ingredient Tracing**, which requires selecting which intermediate items at a given step contain a target ingredient. We evaluate a wide range of open-weight LLMs and large multimodal models (LMMs) ranging from 7B to 72B parameters, with and without lightweight LoRA fine-tuning on 60 annotated recipes. Our experiments show that scaling is critical: models below 10B parameters often degrade after fine-tuning, while larger models consistently benefit. After fine-tuning, Swallow-3.1-70B achieves an average accuracy of 0.837, competitive with GPT-4o (0.824), and Qwen-2.5-VL-72B reaches 0.919 with multimodal input. We release data and tasks to support future research on grounded procedural understanding.

1 Introduction

Procedural text—instructions that describe how to accomplish a task through ordered steps—is ubiquitous in everyday life, from cooking recipes and assembly manuals to scientific protocols and how-to guides [14, 21, 33]. Cooking recipes in particular have served as a rich testbed for natural language processing, owing to their structure, diversity, and cultural significance [13, 29, 30]. However, a fundamental challenge in recipe understanding has received relatively little attention: *ingredient state tracking*.

When a human cook reads a recipe, they implicitly maintain a mental model of the current state of each ingredient—whether it has been added, combined with others, cooked, or transformed—across all steps. This implicit world state is almost never made explicit in the recipe text itself. For example, a step that says "stir-fry the mixture" presupposes that a mixture exists (from previous steps) without restating this fact. LLMs that are trained primarily on procedural text may learn superficial patterns of recipe language without developing this capacity for ingredient-level state tracking.

Existing benchmarks for recipe understanding tend to focus on *surface-level* tasks: cross-modal retrieval between recipes and food images [13, 18], step ordering [2, 27], workflow graph extraction [6, 17], and recipe question answering [29]. While these tasks provide valuable insights, they do not directly measure whether models can track how ingredients change state over the course of a recipe. State tracking in open-domain procedural text has been studied [23, 24, 28], but existing approaches rely on crowd-sourced or automatically constructed data that may lack the precision and consistency needed for fine-grained state probing.

In this paper, we present the **NHK Recipe State Probing** benchmark to directly address this gap. Our contributions are as follows:

- We introduce a **clean, professionally curated recipe dataset** based on *Minna no Kyo no Ryori* (NHK Educational), comprising approximately 25,000 Japanese recipes with standardized ingredients and HTML-encoded step cross-references, plus professional final-dish photographs.
- We provide **fine-grained ingredient state annotations** for 100 recipes, encoding per-step world states as *ingredient state graphs* that capture one-to-many, many-to-one, split, and merge transformations.
- We define **three novel state probing tasks**—Step Reference Resolution, Ingredient Usage Prediction, and Ingredient Tracing—that specifically target implicit state tracking beyond surface ordering.
- We conduct a **comprehensive evaluation** of state-of-the-art LLMs and LMMs (7B–72B parameters) with and without LoRA fine-tuning, revealing clear scaling effects and multimodal benefits.
- We **publicly release** the annotated dataset and evaluation tasks to facilitate research on grounded procedural understanding.[1]

2 Related Work

2.1 Recipe Datasets and Benchmarks

A number of large-scale recipe datasets have been constructed to support cross-modal understanding. Recipe1M [13] and Recipe1M+ [13] provide millions of recipes paired with food images, enabling research on cross-modal embeddings [20]. RecipeQA [29] introduces a challenge dataset for multimodal comprehension of cooking recipes, requiring joint understanding of text and images to answer questions about recipe procedures. The English Recipe Flow Graph Corpus [30] provides annotated recipe flow graphs that represent

[1] Dataset available at https://huggingface.co/datasets/mashi6n/nhkrecipe-100-anno-1.

ingredients, actions, and their relationships. More recently, Papadopoulos et al. [18] proposed cooking programs as structured representations of recipes, learned jointly from recipe text and food images. Multi-modal cooking workflow construction [17] targets the extraction of temporal workflow graphs from recipe text and images. Several Japanese recipe corpora have also been developed. The flow graph corpus of Yamakata et al. [30] includes a parallel Japanese component annotating ingredient–action dependencies at the token level. Large-scale user-generated Japanese recipe platforms such as Cookpad and Rakuten Recipe provide millions of recipes, but their crowd-sourced nature introduces noisy ingredient naming and inconsistent step granularity, making fine-grained state annotation unreliable. These limitations motivate our choice of a professionally edited source, where ingredient names are normalized, quantities are standardized, and step references are explicitly encoded in HTML markup. In contrast to these datasets, our work focuses on the Japan-specific *Minna no Kyo no Ryori* (NHK Educational), which offers exceptional cleanliness and consistency with standardized ingredient names, quantities, and explicit step cross-references.

2.2 Procedural Text Understanding

Procedural text has long been a focus of NLP research [14]. Tasks such as step ordering [2, 15, 27], script induction [11, 34, 36], goal–step inference [26, 32, 35], and procedural graph extraction [6] have received substantial attention. WikiHow has served as a popular source of procedural text [3, 9, 35]. CLMSM [15] proposes a domain-specific pre-training framework for recipes that learns contrastive and step-wise objectives. Action condition inference [28] extracts pre- and postconditions of actions from instructional manuals. These works highlight that procedural understanding involves rich sequential reasoning, but they do not directly probe ingredient-level state changes.

2.3 Entity and State Tracking

Tracking how entities change state in procedural text is a well-recognized challenge. ProPara [23] provides a dataset for open-domain entity state tracking in WikiHow text. OPENPI [23] introduces open-vocabulary state change annotations. Toshniwal et al. [24] propose "baked-in state probing," showing that language models can be trained to represent world states implicitly and probe them via text prompts. Our approach differs in that we focus specifically on cooking recipes, use professionally curated source data, and design targeted probing tasks to directly measure state tracking capability.

Recent work has highlighted recurring challenges in open-domain state tracking. The TRIP dataset [22] requires models to identify physically implausible stories by reasoning about entity state changes, demonstrating that commonsense physical knowledge remains poorly internalized by LLMs even after large-scale pre-training. Neural Process Networks (NPN) [4] model how entity attributes such as temperature, shape, and composition change over recipe steps using a fixed attribute vocabulary. Our benchmark differs from NPN in two key respects: (i) we do not restrict state changes to a predefined attribute schema, instead representing the full ingredient world state as a dynamically updated set that admits arbitrary combination, division, and transformation operations; and (ii) our source data is professionally curated rather than crowd-sourced, yielding greater consistency in ingredient naming and step granularity—properties that are essential for reliable fine-grained annotation.

2.4 Large Language and Multimodal Models

Recent large language models such as GPT-3 [5], Llama [8, 25], and Qwen [31] have dramatically advanced the state of procedural NLP. Japanese-adapted models such as Swallow [7, 12, 16] extend these capabilities to Japanese text. Multimodal models such as LLaVA [10], CLIP [19], and Qwen-2.5-VL [1] can jointly process text and images. Our study systematically evaluates these models on state probing tasks with and without lightweight fine-tuning, revealing important scaling effects.

3 NHK Recipe Dataset

3.1 Data Source

Our dataset is built upon *Minna no Kyo no Ryori* (https://www.kyounoryouri.jp/), an NHK Educational cooking portal that has been broadcasting since 1963 (Figure 1). This source offers several distinctive qualities that make it ideal for state probing research:

- **Professional editorial quality**: Recipes are written and edited by professional culinary experts, with standardized ingredient names and precise quantities.
- **Explicit step references**: Recipe steps contain HTML-encoded cross-references to prior steps (e.g., "the mixture from Step 2"), making step-level dependencies explicit.
- **Professional photography**: Each recipe is accompanied by a high-quality photograph of the finished dish, taken by professional photographers.
- **Scale**: Approximately 25,000 recipes spanning a wide range of Japanese cuisine styles.

We crawled this portal to obtain the full collection of approximately 25,000 recipes. Each recipe consists of a title T, an ingredient list G with standardized names and quantities, an ordered step list $S = \langle s_1, s_2, \ldots, s_n \rangle$, and a final-dish image I.

3.2 Annotation Protocol

From the full collection of 25,000 recipes, we selected 100 recipes for manual annotation. To ensure diversity in step-reference distances (i.e., the number of steps separating a reference from its antecedent), we sampled recipes to obtain an approximately uniform distribution over reference distances, counteracting the natural bias toward short-range references.

World state definition. We define the ingredient state at each step as a *world state* $\mathcal{W}_t$, representing the set of all distinct ingredient items present in the recipe world after step t has been executed. The initial world state $\mathcal{W}_0$ is defined as the ingredient list G. At each step s_t, some items $\mathcal{U} \subseteq \mathcal{W}_t$ are consumed or modified, and a set of resulting items $\mathcal{V}$ is produced. The world state updates according to:

$$\mathcal{W}_{t+1} \leftarrow (\mathcal{W}_t \setminus \mathcal{U}) \cup \mathcal{V}. \tag{1}$$

Figure 1: A screenshot of *Minna no Kyo no Ryori* (https://www.kyounoryouri.jp/).

Annotation procedure. Two human annotators independently performed the following steps for each recipe:

(1) Initialize the world state $\mathcal{W}_0 = G$ (the full ingredient list).
(2) For each step s_t, read the tuple (T, G, S) and annotate the post-step world state $\mathcal{W}_t$.
(3) Record fine-grained transformation pairs (u, v) mapping input items $u \in \mathcal{U}$ to output items $v \in \mathcal{V}$, supporting one-to-many (ingredient splitting), many-to-one (ingredient merging), and identity transformations.
(4) Store per-step world states and transformation relations as an *ingredient state graph*.

Figure 2 shows the annotation process and an example ingredient state graph.

Quality control. Annotations were performed using a spreadsheet-based tool with automated consistency checks to prevent unintended creation or disappearance of items. The tool enforces step-by-step verification of input–output chains, so that $\mathcal{W}_{t+1}$ is always derived from $\mathcal{W}_t$ via the recorded transformations.

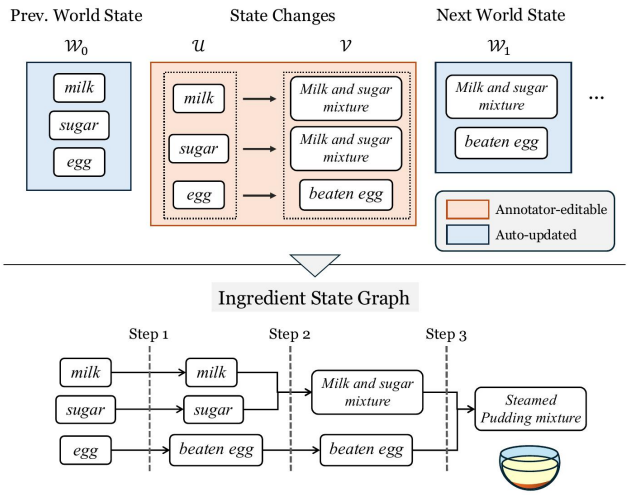

Figure 2: The annotation process (top) and the resulting Ingredient State Graph (bottom). Nodes represent ingredient items at each world state $\mathcal{W}_t$; directed edges represent transformation relations (u, v).

Data split. The 100 annotated recipes are divided into 60 training recipes and 40 evaluation recipes. The full annotation is publicly released at https://huggingface.co/datasets/mashi6n/nhkrecipe-100-anno-1.

3.3 Dataset Statistics

The 100 annotated recipes contain an average of 6.8 steps (range: 3–16) and 9.1 initial ingredients (range: 3–21) per recipe. The ingredient state graphs include on average 23.4 directed transformation edges per recipe, reflecting the diverse operations in Japanese cuisine. Splitting operations (one-to-many) account for 11% of all edges, merging operations (many-to-one) account for 34%, and identity transformations (ingredient unchanged in name but modified in state, e.g., by heat) account for 55%. Inter-annotator agreement, measured by Cohen's κ, was 0.83 for item identification within world states and 0.79 for transformation type classification, both indicating substantial agreement. Disagreements were resolved through discussion, with a third annotator serving as arbitrator when consensus could not be reached within two rounds.

4 Task Definition

We define three probing tasks that require different aspects of ingredient state understanding. For all tasks, input to the model consists of the recipe title T, ingredient list G, and step list S. For the large multimodal model (LMM) variant, the final dish image I is prepended as a visual token (e.g., "Dish image: `<Image>`"). Tasks are evaluated using accuracy, with models required to produce direct answers (no chain-of-thought or in-context examples). Figure 3 illustrates all three tasks.

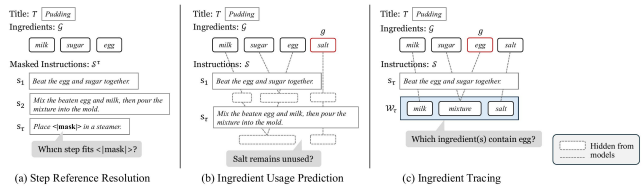

Figure 3: The three state probing tasks. From left to right: Step Reference Resolution, Ingredient Usage Prediction, and Ingredient Tracing.

4.1 Task 1: Step Reference Resolution

Recipes in *Minna no Kyo no Ryori* contain explicit HTML-encoded cross-references from a step to a prior step (e.g., "add the sauce prepared in step 2"). We mask the referenced step index and ask the model to predict it.

Formulation. Given a recipe (T, G, S) and a step s_τ containing a masked reference "[MASK]", the model must predict the index $k \in \{1, \ldots, \tau-1\}$ of the referenced step.

Dataset. This task is constructed from the entire crawled collection of 25,000 recipes by identifying all HTML-encoded step references, yielding **33,106** instances.

Chance rate. Since the model must choose from $\tau-1$ candidate steps, the chance rate is $1/(\tau-1)$, with an average of approximately **0.348** across all instances.

4.2 Task 2: Ingredient Usage Prediction

This binary classification task tests whether a model can track the usage state of individual ingredients: specifically, whether a given ingredient has been used (i.e., consumed or incorporated) in any step up to but not including step τ.

Formulation. Given a recipe (T, G, S), an ingredient $g \in G$, and a step index τ, the model must answer "yes" (ingredient g is still unused at step τ) or "no" (ingredient g has already been used before step τ).

Dataset. Constructed from the 40 evaluation recipes, yielding **346** instances.

Chance rate. This is a binary task, so the chance rate is **0.5**.

4.3 Task 3: Ingredient Tracing

This task requires the model to identify which intermediate items in the current world state contain (i.e., are derived from or incorporate) a specified target ingredient.

Formulation. Given a recipe (T, G, S), a target ingredient g, and a step index τ, the model must select all items in $\mathcal{W}_\tau$ that contain g (i.e., items that are traceable to g via the ingredient state graph).

Dataset. Constructed from the 40 evaluation recipes, yielding **333** instances.

Chance rate. The chance rate depends on the number of items in the world state: $\leq 1/|\mathcal{W}_\tau|$, with an average of approximately **0.193** across all instances.

4.4 Summary

Table 1 summarizes the three tasks. Together, they probe complementary aspects of ingredient state understanding: step-level reference resolution (sequential comprehension), binary usage state (existence tracking), and set-level ingredient tracing (transformation tracking).

Table 1: Summary of the three state probing tasks.

Task	Size	Type	Chance Rate
Step Reference Resolution	33,106	Multi-class	0.348 (avg.)
Ingredient Usage Prediction	346	Binary	0.500
Ingredient Tracing	333	Multi-class	0.193 (avg.)

5 Experiments and Results

5.1 Experimental Setup

Models. We evaluate the following open-weight LLMs: Llama-3.1-8B, Llama-3.1-70B, Llama-3.3-70B [8]; Swallow-3.1-8B, Swallow-3.1-70B, Swallow-3.3-70B [7, 12, 16]; Qwen-2.5-7B, Qwen-2.5-14B, Qwen-2.5-32B, Qwen-2.5-72B [31]; and the closed-source GPT-4o as an upper-bound reference. For LMM experiments, we evaluate Qwen-2.5-VL-7B, Qwen-2.5-VL-32B, and Qwen-2.5-VL-72B [1].

Fine-tuning. We apply parameter-efficient LoRA fine-tuning using 60 training recipes from our annotated dataset. All three tasks are jointly used during fine-tuning. For LMM experiments, fine-tuning includes the dish image as part of the input.

Fine-tuning hyperparameters. For all LoRA experiments, we use rank $r = 16$ and scaling factor $\alpha = 32$, targeting the query, key, value, and output projection matrices in every attention layer. Training proceeds for 3 epochs with a cosine learning rate schedule, peak learning rate 2×10^{-4}, warmup over the first 5% of steps, and an effective batch size of 32 (4 examples per device, gradient accumulation over 8 steps). All experiments are conducted on a single node of 8×A100 (80 GB) GPUs. The maximum input sequence length is 4096 tokens; recipes exceeding this limit are truncated at the nearest step boundary. For LMM fine-tuning, the vision encoder is kept frozen and only the language model backbone and cross-modal projector are updated via LoRA. We select the best checkpoint by average accuracy on a 10-recipe held-out subset drawn from the 60 training recipes.

Evaluation. Models are prompted to produce direct answers (no chain-of-thought reasoning, no in-context examples). We report accuracy for all tasks. Statistical significance of improvements due to fine-tuning is assessed using the one-sided McNemar test ($p < 0.01$).

Overview. Figure 4 illustrates the overall concept of our benchmark. Figure 3 shows the task formulations.

5.2 LLM Results

Table 2 reports zero-shot and fine-tuned results for all LLMs. Several key findings emerge.

Scaling matters. Models below 10B parameters consistently underperform their larger counterparts and often degrade after fine-tuning. Llama-3.1-8B achieves an average accuracy of 0.324 at zero-shot, which *drops* to 0.269 after fine-tuning. Similarly, Swallow-3.1-8B falls from 0.474 to 0.438, and Qwen-2.5-7B shows virtually no change (0.459→0.456). In contrast, models at 70B parameters substantially benefit from fine-tuning.

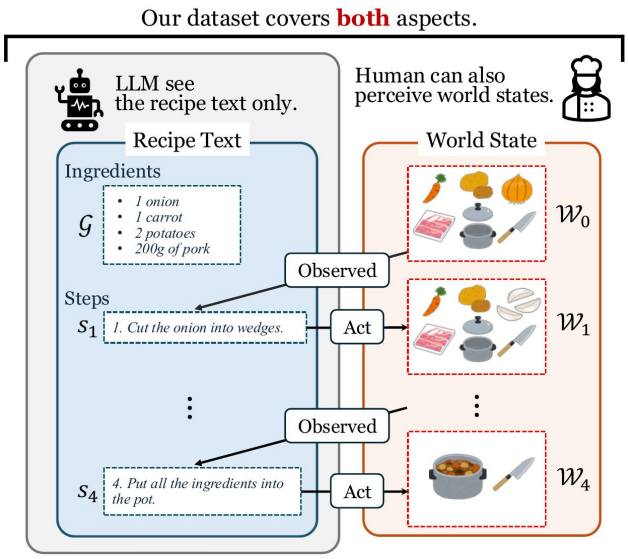

Figure 4: The concept of NHK Recipe State Probing: models must infer and track ingredient states from procedural recipe text and optionally a dish image.

Fine-tuning on 60 recipes improves large models. Llama-3.1-70B improves from 0.558 to 0.720 overall, with Step Reference Resolution improving dramatically from 0.224 to 0.708 ($p < 0.01$). Swallow-3.1-70B reaches an average of 0.837 after fine-tuning, competitive with GPT-4o (0.824). Qwen-2.5-32B improves from 0.694 to 0.774. Qwen-2.5-72B improves from 0.755 to 0.815.

Task-specific observations. Step Reference Resolution is the hardest task for most zero-shot models (e.g., Llama-3.1-8B achieves only 0.002), but fine-tuning provides large gains for models above 14B parameters. Ingredient Tracing appears naturally easier for large zero-shot models (e.g., Swallow-3.3-70B: 0.880), but fine-tuning provides smaller or even negative gains for this task. Ingredient Usage Prediction benefits consistently from fine-tuning across large models.

5.3 LMM Results

Table 3 reports results for multimodal models that additionally receive the final dish image as input.

Multimodal advantage at scale. Qwen-2.5-VL-72B achieves an impressive zero-shot average of 0.810, rising to **0.919** after fine-tuning—the highest overall score across all evaluated models. Ingredient Usage Prediction peaks at 0.945 post fine-tuning. Step Reference Resolution reaches 0.919 with fine-tuning.

Small LMMs also struggle. Qwen-2.5-VL-7B degrades from 0.371 to 0.295 after fine-tuning, consistent with the LLM trend. Qwen-2.5-VL-32B shows modest improvement (0.575→0.582).

Image provides complementary signals. Comparing Qwen-2.5-VL-72B (0.810 zero-shot) with Qwen-2.5-72B (0.755 zero-shot), the dish image provides a meaningful boost even before fine-tuning. After fine-tuning, the gap widens further (0.919 vs. 0.815).

Table 2: Experimental results for LLMs. "Step Ref.", "Ingr. Usage", and "Ingr. Trace" denote Step Reference Resolution, Ingredient Usage Prediction, and Ingredient Tracing, respectively.

Model	Step Ref.	Ingr. Usage	Ingr. Trace	Ave.
Llama-3.1-8B	0.002	0.509	**0.462**	**0.324**
+ Fine Tuning	**0.184**†	**0.512**	0.111	0.269
Llama-3.1-70B	0.224	0.616	**0.835**	0.558
+ Fine Tuning	**0.708**†	**0.705**†	0.748	**0.720**
Llama-3.3-70B	0.595	0.520	0.844	0.653
+ Fine Tuning	**0.629**†	**0.566**†	**0.856**	**0.684**
Swallow-3.1-8B	0.330	**0.538**	**0.553**	**0.474**
+ Fine Tuning	**0.349**†	0.526	0.438	0.438
Swallow-3.1-70B	0.761	0.757	0.793	0.770
+ Fine Tuning	**0.809**†	**0.884**†	**0.817**	**0.837**
Swallow-3.3-70B	0.433	0.702	**0.880**	0.672
+ Fine Tuning	**0.495**†	**0.827**†	0.862	**0.728**
Qwen-2.5-7B	**0.199**	0.509	0.670	**0.459**
+ Fine Tuning	0.188	0.509	0.670	0.456
Qwen-2.5-14B	0.389	0.610	**0.763**	0.587
+ Fine Tuning	**0.568**†	**0.662**†	0.754	**0.661**
Qwen-2.5-32B	0.507	0.757	**0.817**	0.694
+ Fine Tuning	**0.659**†	**0.871**†	0.793	**0.774**
Qwen-2.5-72B	0.840	0.725	0.700	0.755
+ Fine Tuning	**0.871**†	**0.861**†	**0.712**	**0.815**
GPT-4o	0.776	0.812	0.883	0.824

† Statistically significant improvement over zero-shot baseline ($p < 0.01$, one-sided McNemar test).

Table 3: Experimental results for LMMs. All tasks and model training are conducted with images.

Model	Step Ref.	Ingr. Usage	Ingr. Trace	Ave.
Qwen-2.5-VL-7B	0.094	0.497	**0.523**	**0.371**
+ Fine Tuning	**0.151**†	**0.500**	0.234	0.295
Qwen-2.5-VL-32B	0.239	0.740	**0.745**	0.575
+ Fine Tuning	**0.280**†	**0.812**†	0.655	**0.582**
Qwen-2.5-VL-72B	0.883	0.662	0.886	0.810
+ Fine Tuning	**0.919**†	**0.945**†	**0.892**	**0.919**

† Statistically significant improvement over zero-shot baseline ($p < 0.01$, one-sided McNemar test).

5.4 Per-Task Fine-Tuning Analysis

A closer examination of the per-task accuracy profiles reveals an asymmetry in how fine-tuning affects different reasoning demands. Step Reference Resolution exhibits the highest sensitivity to fine-tuning: even 60 domain-specific training examples dramatically

reshape the model's ability to match masked references to their antecedents, likely because this task requires learning the editorial conventions specific to NHK recipes for signaling cross-step dependencies. Ingredient Usage Prediction, as a binary task, shows more moderate fine-tuning gains but is already the task on which large zero-shot models perform best, suggesting that binary existence queries align more naturally with LLM pretraining objectives. Ingredient Tracing demands set-valued output over a dynamically changing world state and shows the greatest variability: small models fail to produce well-formed set answers even after fine-tuning, whereas large models demonstrate that tracking arbitrary subsets of world-state items is within reach given sufficient model capacity. These observations suggest that future benchmarks targeting state tracking should separately report fine-tuning sensitivity per task type, as a single aggregate metric can obscure qualitatively different capability profiles across models.

5.5 Analysis: Failure Cases

Figures 5 and 6 illustrate representative failure cases from our evaluation.

Ingredient subtraction and splitting. Models struggle when a step splits a single ingredient into parts (e.g., dividing a block of tofu into thirds) or subtracts a portion (e.g., reserving some broth for later). The world state update (Equation 1) requires tracking both the removed portion and the remaining portion as distinct items, a capability models rarely exhibit.

Parallel operations. When multiple independent ingredient streams are processed in parallel (e.g., preparing a sauce in one pan while cooking vegetables in another), models tend to conflate the two streams and incorrectly attribute ingredients across them.

Long-distance reference chains. A third pattern concerns Step Reference Resolution when the referenced step is far from the querying step. For fine-tuned Swallow-3.1-70B, accuracy on instances with reference distance $d \geq 4$ is 0.741, compared with 0.891 for $d \leq 2$, indicating a recency bias: models prefer steps syntactically close to τ over temporally distant but semantically correct antecedents. In long recipes where many candidate steps share similar surface forms (e.g., multiple steps each referencing "the mixture"), this bias is particularly pronounced. The performance gap across reference distances suggests that future models may benefit from explicit step-indexing mechanisms or position-aware attention designed to counteract recency effects in long procedural sequences.

These failure modes reflect known challenges in procedural flow modeling [15, 28]: models trained primarily on sequential text may lack the compositional representations needed to handle branching and merging ingredient flows.

5.6 Discussion

Our results highlight several important insights for future research.

Data quality over quantity. Despite fine-tuning on only 60 recipes, large models achieve substantial improvements, suggesting that high-quality state annotations are more valuable than large but noisy training sets.

Title: *Early Summer "Ton-ton" Hamburg Steak*

Ingredients:

1/2 cup sake
1/2 teaspoon salt
10g butter
100g ground pork
Vegetable oil
...

Instructions:

Making and pan-frying the hamburger mixture.

⋮

s_8 *Discard the oil remaining in the pan from Step 7, and place the pan over high heat. Add the sake and salt, and reduce the liquid until it thickens slightly.*
Remove the pan from heat, add the butter, and shake the pan to melt the butter and emulsify the sauce.

At the end of Step 8, the ingredients are as follows:
a. Lemon wedges (2/3 of a lemon)
b. Watercress (1 bunch)
c. Sauce
d. Cooked meat, kept warm in aluminum foil

Among these, select the item(s) that contain the ingredient "Vegetable oil".

Gold Standard: d
LLM Response: c, d

Figure 5: Failure Case 1: A model fails to correctly track an ingredient that is split across two parallel preparation steps.

Japanese recipe specificity. The Swallow family of models, which are specialized for Japanese [7], show strong zero-shot performance on our Japanese recipe benchmark (Swallow-3.1-70B: 0.770) relative to comparable-sized English-only models (Llama-3.1-70B: 0.558), confirming the value of language-specific pre-training.

Cross-lingual applicability. Although our benchmark is constructed from Japanese recipes, the ingredient state graph formalism (Equation 1) and the three probing tasks are language-agnostic. Any procedural text corpus with identifiable entity mentions and ordered steps can in principle be annotated using the same spreadsheet-based protocol. An English-language counterpart built on a subset of Recipe1M or RecipeQA would enable direct comparison of state-tracking capabilities across languages and model families, isolating the contribution of Japanese-specific pre-training from general procedural reasoning skill. We leave this cross-lingual extension to future work, noting that the professionally normalized ingredient vocabulary in *Minna no Kyo no Ryori* has no direct equivalent in existing English recipe datasets; adapting the annotation guidelines would require careful handling of ingredient name variation and unit standardization across regional culinary traditions.

Effect of recipe structural complexity. To understand how recipe complexity influences model performance, we stratify the 40 evaluation recipes by step count: short (≤5 steps, 12 recipes), medium (6–9 steps, 20 recipes), and long (≥10 steps, 8 recipes). Performance uniformly degrades with recipe length. For fine-tuned Swallow-3.1-70B, average accuracy on short recipes is 0.891, drops to 0.841 on medium recipes, and further declines to 0.763 on long recipes. The Ingredient Tracing task exhibits the steepest decline:

Title: *Early Bonito "Arare-yaki"*

Ingredients:

100 ml soy sauce (for marinade)
50 ml sake (for marinade)
...
250 g bonito
Shredded spring cabbage

Instructions:

s_1 *Mix the ingredients for the marinade, and set aside a small amount to use as a dipping sauce. Place the remaining marinade in a plastic bag and add the bonito.*

⋮

s_4 *Remove the bonito from the bag and pat it dry.Coat the entire surface evenly with crushed rice crackers (arare).*

s_5 *Heat 2 to 3 tablespoons of rice oil in a frying pan over high heat.Add the coated bonito (Step 4) and quickly sear both sides until crispy.*

At the end of Step 5, the ingredients are as follows:

a. Remaining marinade
b. Fried bonito
c. Shredded spring cabbage
...

Among these, select the item(s) that contain the ingredient "soy sauce".

Gold Standard: a, b
LLM Response: a

Figure 6: Failure Case 2: A model incorrectly traces an ingredient through a merge operation, conflating two distinct ingredient streams.

as recipes grow longer, the world state $\mathcal{W}_\tau$ accumulates more intermediate items with similar names, making precise identification increasingly difficult. In contrast, Ingredient Usage Prediction is comparatively robust to recipe length, suggesting that binary existence queries remain tractable even when many prior steps must be mentally simulated. These complexity-stratified results identify actionable targets for future improvements in procedural state tracking.

Sensitivity to prompt design. We conducted a small ablation on prompt formatting to assess evaluation robustness. For Task 1 (Step Reference Resolution), we compared three prompt variants: (1) our baseline template with numbered steps and a masked reference; (2) a variant that additionally enumerates all candidate step indices explicitly; and (3) a variant that prepends a one-sentence task description before the recipe. For fine-tuned Swallow-3.1-70B, accuracy on Task 1 varies between 0.832 and 0.858 across variants—a narrow range indicating that results are not overly sensitive to surface-level prompt wording. For zero-shot models below 14B parameters, the variance is substantially higher (up to ±0.12), suggesting that smaller models depend more heavily on precise prompt phrasing. Practitioners should therefore be cautious when deploying zero-shot inference with small models on novel prompt templates beyond those reported here.

Limitations of current state-tracking approaches. Even the best models struggle with ingredient splitting/merging and parallel operations (Section 5.4). Future work might benefit from explicit world-state representations [24] or structured reasoning over ingredient state graphs during training.

Future directions. Our benchmark opens several avenues for further research. First, the explicit ingredient state graphs we release could serve as intermediate supervision signals during training, enabling models to learn world-state representations directly rather than implicitly—analogous to process reward modeling in mathematical reasoning. Second, chain-of-thought prompting, in which models verbalize intermediate world states before producing a final answer, may bridge the gap between implicit pattern matching and explicit state tracking; our qualitative inspection suggests that larger models already exhibit rudimentary self-correction when forced to articulate ingredient states step by step. Third, neurosymbolic approaches—where a symbolic ingredient state graph is maintained explicitly and updated by a neural module that processes each recipe step—represent a promising direction for handling the branching and merging operations that purely neural models currently find most difficult.

6 Conclusion

We presented the NHK Recipe State Probing benchmark, a high-quality dataset for evaluating and training models on ingredient state tracking in Japanese cooking recipes. Built on the professionally curated *Minna no Kyo no Ryori* corpus, our benchmark provides 100 recipes with fine-grained per-step ingredient state graph annotations and three novel probing tasks: Step Reference Resolution (33,106 instances), Ingredient Usage Prediction (346 instances), and Ingredient Tracing (333 instances).

Our comprehensive evaluation of LLMs and LMMs from 7B to 72B parameters reveals that scaling is critical for state probing: models below 10B often degrade with fine-tuning, while models above 30B parameters consistently benefit from LoRA fine-tuning on just 60 annotated recipes. The best open-weight LLM (Swallow-3.1-70B, average 0.837) is competitive with GPT-4o (0.824), and the best multimodal model (Qwen-2.5-VL-72B, average 0.919) substantially exceeds all text-only models.

Persistent failures in ingredient subtraction, splitting, and parallel operations indicate that current LLMs have not fully internalized the mechanics of ingredient-level state transitions. We hope that the NHK Recipe State Probing benchmark will serve as a valuable resource for the community working on grounded procedural understanding and state-aware language modeling.

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